Experimental Results: PIGNN-Attn-LS across Power System Cases

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1 Experimental Setup

1.1 Baseline Definition

The **baseline** throughout all experiments is a *flat-start* initialisation: every bus voltage magnitude is set to $|V| = 1.0$ p.u. and every voltage angle to $\theta = 0°$. This is the standard cold-start point used in classical Newton–Raphson solvers when no better estimate is available. The baseline RMSE and angle error reported below are the errors of this flat-start guess against the Newton–Raphson converged solution — *before* the model does any work. The model's job is to reduce these errors.

1.2 Test Cases: Bus and Branch Counts

Table 1 lists the N_{bus} and N_{branch} for each test case. The branch count determines the number of edges in the graph and directly affects the cost of the edge-aware attention.

Table 1: Grid size of test cases.

Case	N_{bus}	N_{branch}	Notes
case14_ppcY	14	20	IEEE 14-bus
case24_ieee_rts	24	38	IEEE RTS-24
case30_ppcY_A	30	41	IEEE 30-bus
case33bw_ppcY_A	33	32	Radial distribution
case39_ppcY_A	39	46	New England 39-bus
case57_ppcY_A	57	80	IEEE 57-bus
case89pegase	89	210	PEGASE 89-bus (dense)
case118_ppcY_A	118	186	IEEE 118-bus
case145_ppcY_A	145	453	PEGASE 145-bus (very dense)

2 Full Baseline vs. Model Comparison

Table 2 summarises the RMSE of the flat-start baseline and the trained model across all cases. All model runs use $K = 40$, $d = 4$, $d_{\text{hi}} = 16$, `numattn`$= 1$, Armijo$=$ True, 100 epochs, train ratio $= 1/3$.

Reading the baseline. The flat-start error is *not* zero because real grids are not at uniform $1.0\angle 0°$ p.u. at operating conditions. The deviation grows with grid complexity: case145 has a flat-start angle error of $18.64°$ because the PEGASE 145-bus network has strong reactive-power variation and several high-angle buses. case57 has a large flat-start $|V|$ error (0.126 p.u.) because it contains several buses with voltages that differ considerably from 1.0 p.u. at the operating point.

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Table 2: Flat-start baseline vs. PIGNN-Attn-LS model. Reductions are $\lfloor \text{baseline} / \text{model} \rfloor$.

Case	N_{bus}	$\lvert V \rvert$ RMSE (p.u.)		Reduc.	θ RMSE (°)		Reduc.
		Baseline	Model		Baseline	Model	
case24_ieee_rts	24	2.24×10^{-2}	1×10^{-5}	$\sim 2000\times$	0.91	0.01	$\sim 90\times$
case30_ppcY_A	30	1.83×10^{-2}	7×10^{-5}	$\sim 260\times$	0.21	0.025	$\sim 8\times$
case33bw_ppcY_A	33	5.96×10^{-2}	5×10^{-4}	$\sim 120\times$	1.47	0.10	$\sim 15\times$
case39_ppcY_A	39	1.41×10^{-2}	5×10^{-5}	$\sim 280\times$	1.50	0.07	$\sim 21\times$
case57_ppcY_A	57	1.26×10^{-1}	3×10^{-3}	$\sim 42\times$	1.13	0.30	$\sim 4\times$
case89pegase	89	2.03×10^{-2}	2.5×10^{-3}	$\sim 8\times$	0.30	0.27	$\sim 1.1\times$
case118_ppcY_A	118	9.02×10^{-3}	6×10^{-5}	$\sim 150\times$	2.73	0.08	$\sim 34\times$
case145_ppcY_A	145	6.52×10^{-2}	1.77×10^{-3}	$\sim 37\times$	18.64	1.77	$\sim 11\times$

3 Armijo vs. Non-Armijo

The backtracking Armijo line search selects a step size $\alpha \in \{1, \rho, \rho^2, \ldots\}$ at each unrolled step that satisfies the sufficient-decrease condition on the power-mismatch ℓ^∞-norm:

$$F(x + \alpha\, \Delta x) \;\leq\; (1 - c_1\, \alpha)\, F(x).$$

Without Armijo (`armijo=False`), the full proposed step $\alpha = 1$ is always accepted, which can cause oscillation or divergence when the update overshoots. Table 3 shows the effect across cases.

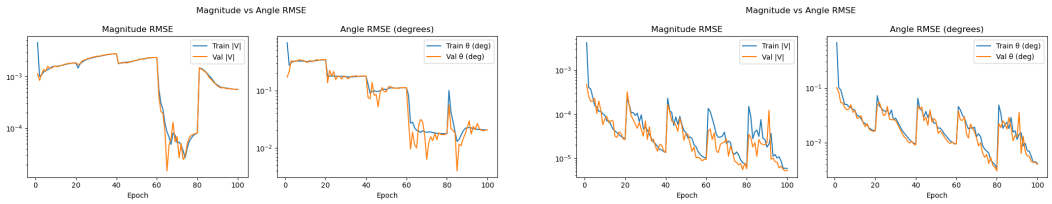
Table 3: Effect of Armijo line search. $K = 40$, $d = 4$, $d_{\text{hi}} = 16$, `numattn`$= 1$, 100 ep.

Case	Without Armijo		With Armijo		Verdict
	$\lvert V \rvert$ RMSE	θ (°)	$\lvert V \rvert$ RMSE	θ (°)	
case24_ieee_rts	6×10^{-4}	0.025	$\mathbf{1 \times 10^{-5}}$	**0.01**	Large gain
case30_ppcY_A	3×10^{-4}	0.20	$\mathbf{7 \times 10^{-5}}$	**0.025**	Large gain
case33bw_ppcY_A	1.5×10^{-3}	0.10	5×10^{-4}	0.10	Modest gain
case39_ppcY_A	5×10^{-3}	0.10	$\mathbf{5 \times 10^{-5}}$	**0.07**	Large gain
case57_ppcY_A	3×10^{-3}	0.20	3×10^{-3}	0.30	No gain
case89pegase	3×10^{-3}	0.30	2.5×10^{-3}	0.27	Marginal
case118_ppcY_A	4×10^{-4}	0.09	$\mathbf{6 \times 10^{-5}}$	**0.08**	Large gain
case145_ppcY_A	0.010	6.0	3×10^{-3}	6.0	Modest gain

For most cases, Armijo produces a significant accuracy improvement by preventing divergent updates. The exceptions are case57 and case89pegase, which appear to be architecturally undersized at ($K = 40$, $d_{\text{hi}} = 16$): both cases plateau regardless of the step-size policy. case89pegase is notably dense ($N_{\text{branch}} = 210$ for 89 buses), which may require a larger model.

3.1 Representative Learning Curves

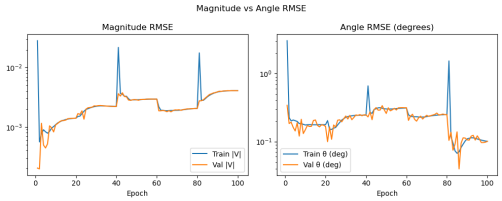
Figures 1–4 show the validation RMSE curves ($|V|$ and θ) for four representative cases, contrasting Armijo=False and Armijo=True.



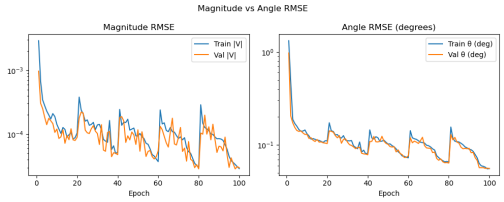
(a) case24 – Armijo=False

(b) case24 – Armijo=True

Figure 1: case24_ieee_rts: Armijo reduces $|V|$ RMSE by $\sim 60\times$ and stabilises training.

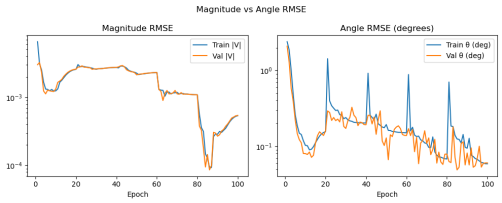


(a) case39 – Armijo=False (diverges)

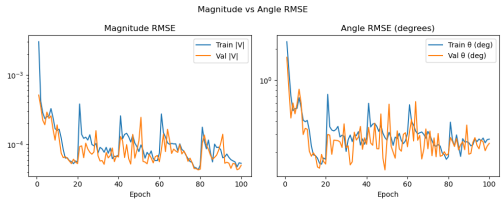


(b) case39 – Armijo=True (converges)

Figure 2: case39_ppcY_A: without Armijo the updates overshoot and diverge; Armijo restores stable descent.



(a) case118 – Armijo=False



(b) case118 – Armijo=True

Figure 3: case118_ppcY_A: Armijo reduces $|V|$ RMSE by $\sim 7\times$ and produces a sawtooth descent pattern consistent with cosine-restart scheduling.

4 case145 Architecture Sweep

4.1 What is the Pareto Front?

In a multi-objective setting, a run is **Pareto-optimal** (or Pareto-dominant) if no other run is *strictly better on every objective simultaneously*. Here the two objectives are:

1. Minimise $|V|$ RMSE (voltage magnitude accuracy).
2. Minimise θ RMSE (angle accuracy).

A run is on the Pareto front if you cannot improve one metric without making the other worse. Pareto-optimal runs represent the *accuracy–accuracy* trade-off frontier and are the natural candidates for deployment depending on which metric the operator prioritises.

4.2 Part A: Launcher Log Sweep

All runs: $K = 40$, 100 epochs, Armijo= False. Sorted by θ RMSE (best first). Runs with $d_{\text{hi}} = 64$ diverged catastrophically (physics loss $\sim 10^5$) and are listed separately. The OOM run is excluded.

The three Pareto-optimal runs are:

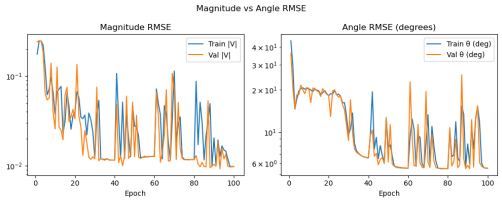
- **Rank 1** ($H = 8$, $d_{\text{hi}} = 24$, $L = 8$, 65k params): best θ at $1.77°$.
- **Rank 5** ($H = 12$, $d_{\text{hi}} = 24$, $L = 4$, 36k params): best $|V|$ RMSE at 1.21×10^{-3}, smallest Pareto model.
- **Rank 9** ($H = 8$, $d_{\text{hi}} = 24$, $L = 4$, 36k params): good $|V|$ RMSE (1.67×10^{-3}) at minimal cost; also 36k params but worse than Rank 5 on $|V|$.

4.3 Part B: Earlier Plot-Based Runs (small configs, numattn $= 1$–2)

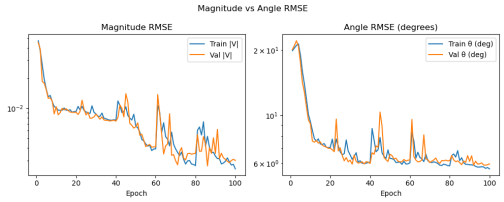
These runs predate the full sweep and used smaller d_{hi} and fewer attention layers.

4.4 Key Findings

1. **Best overall:** H=8, d_{hi}=24, L=8 (65k params) achieves $\theta = 1.77°$, $|V| = 1.77 \times 10^{-3}$.



(a) case145 – Armijo=False (oscillates)



(b) case145 – Armijo=True (smoother)

Figure 4: case145_ppcY_A (small config, $d_{\text{hi}} = 16$, `numattn`$= 1$): Armijo reduces $|V|$ RMSE by $\sim 3\times$ but θ RMSE stays near $6°$, indicating the model is undersized for this case.

Table 4: case145 architecture sweep (launcher log sweep). ✓ = Pareto-optimal. H=heads, L=attn layers.

Rank	Label	H	d_{hi}	L	Params	$\mathcal{L}_{\text{phys}}$	$\lvert V\rvert$ RMSE	θ (°)	ΔP^{∞}	Pareto
1	03_nhead8_dhi24_attn8	8	24	8	65,896	0.426	1.767×10^{-3}	**1.770**	2.750	✓
2	09_nhead8_dhi32_attn10	8	32	10	137,288	0.498	2.758×10^{-3}	2.440	**0.320**	
3	04_nhead8_dhi24_attn10	8	24	10	80,816	0.461	4.140×10^{-3}	2.776	0.457	
4	08_nhead8_dhi32_attn8	8	32	8	111,472	0.458	6.323×10^{-3}	2.824	0.990	
5	16_nhead12_dhi24_attn4	12	24	4	36,360	0.503	$\mathbf{1.213 \times 10^{-3}}$	3.973	3.374	✓
6	17_nhead12_dhi24_attn6	12	24	6	51,432	0.461	5.496×10^{-3}	4.012	5.520	
7	05_nhead8_dhi24_attn12	8	24	12	95,736	0.431	3.472×10^{-2}	4.075	1.778	
8	18_nhead12_dhi24_attn8	12	24	8	66,504	0.836	2.324×10^{-2}	4.794	1.114	
9	01_nhead8_dhi24_attn4	8	24	4	36,056	0.473	1.671×10^{-3}	4.945	1.148	✓
10	06_nhead8_dhi32_attn4	8	32	4	59,840	0.493	2.622×10^{-2}	5.079	1.350	
11	02_nhead8_dhi24_attn6	8	24	6	50,976	0.446	4.395×10^{-3}	5.411	4.612	
12	07_nhead8_dhi32_attn6	8	32	6	85,656	0.513	3.259×10^{-3}	6.354	9.195	
OOM	10_nhead8_dhi32_attn12	8	32	12	163,104	out of memory				
FAIL	11_nhead8_dhi64_attn4	8	64	4	216,416	1.51×10^{5}	0.246	39.6°	490	diverged
FAIL	12_nhead8_dhi64_attn6	8	64	6	316,536	1.66×10^{5}	0.241	88.0°	729	diverged
—	runs 19–45	nhead=12 ($d_{\text{hi}} \geq 48$), nhead=16 – incomplete								

2. **Best $|V|$ RMSE:** H=12, d_{hi}=24, L=4 (36k params) achieves $|V| = 1.21 \times 10^{-3}$, but $\theta = 3.97°$.
3. **Deeper is not always better:** for $d_{\text{hi}} = 24$, L=8 outperforms both L=10 and L=12 on angle accuracy. L=12 has $3\times$ the angle RMSE of L=8.
4. $d_{\text{hi}} = 64$ **diverged completely:** physics loss jumped to $\sim 10^5$, θ errors of 40–88°. The likely cause is that the large hidden dimension magnifies gradient variance at initialisation (all heads initialised to zero), making the early training unstable under the physics loss only.
5. **Small configs plateau at $\theta \approx$ 5–6°:** the earlier runs with $d_{\text{hi}} \leq 16$ and `numattn`≤ 2 all converge to this plateau, confirming that architectural capacity (not just steps K) is the bottleneck for case145.
6. **Many sweep runs still incomplete:** nhead$= 12$ with $d_{\text{hi}} \geq 48$ and all nhead$= 16$ runs are pending; better configurations may yet be found.

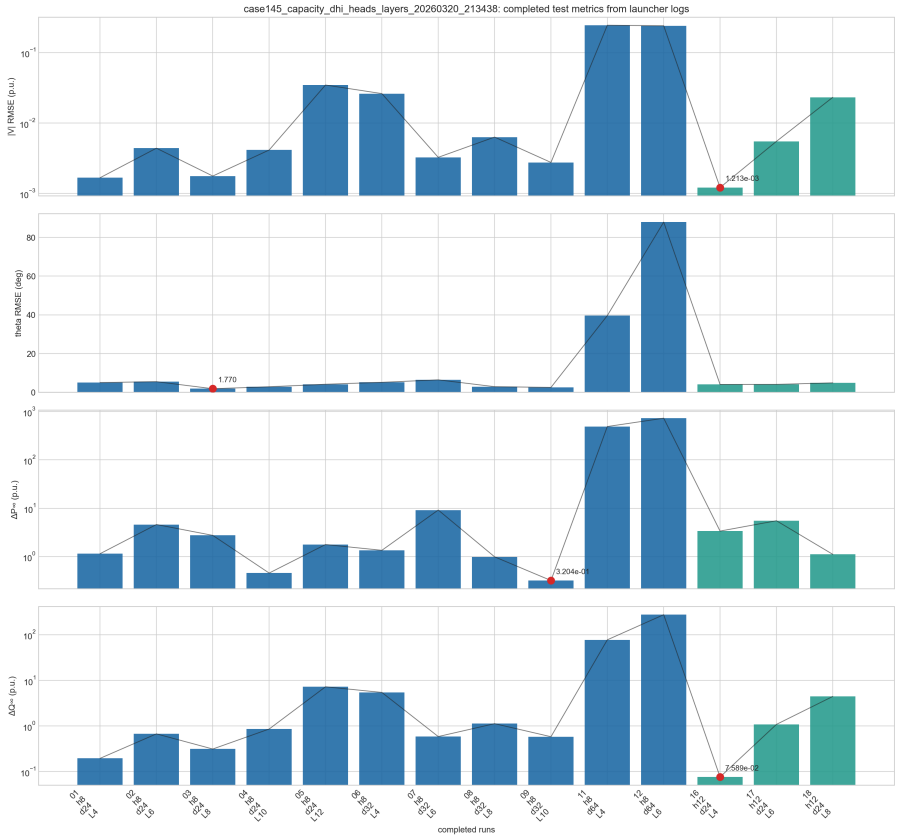


Figure 5: case145 launcher sweep: test metrics by run ($K = 40$, Armijo=False). Pareto-optimal runs are the best in the lower-left of the $|V|$–θ plane.

Table 5: case145 earlier runs (from validation learning curves).

Config	K	d	d_{hi}	`numattn`	Armijo	$\lvert V\rvert$ RMSE	θ (°)
K40_d4_dhi16_n1	40	4	16	1	False	~ 0.010	~ 6
K40_d4_dhi16_n1	40	4	16	1	**True**	$\sim 3 \times 10^{-3}$	~ 6
K40_d4_dhi16_n2	40	4	16	2	False	$\sim 2 \times 10^{-3}$	~ 5
K40_d4_dhi24_n1	40	4	24	1	False	~ 0.010	~ 5
K40_d4_dhi24_n2	40	4	24	2	False	$\sim 8 \times 10^{-3}$	~ 5
K40_d6_dhi16_n1	40	6	16	1	False	~ 0.030	~ 6
K40_d6_dhi24_n1	40	6	24	1	False	~ 0.015	~ 5
K48_d4_dhi16_n1	48	4	16	1	False	$\sim 3 \times 10^{-3}$	~ 5
K48_d4_dhi24_n2	48	4	24	2	False	$\sim 3 \times 10^{-3}$	~ 5
K48_d6_dhi24_n2	48	6	24	2	False	~ 0.010	~ 10

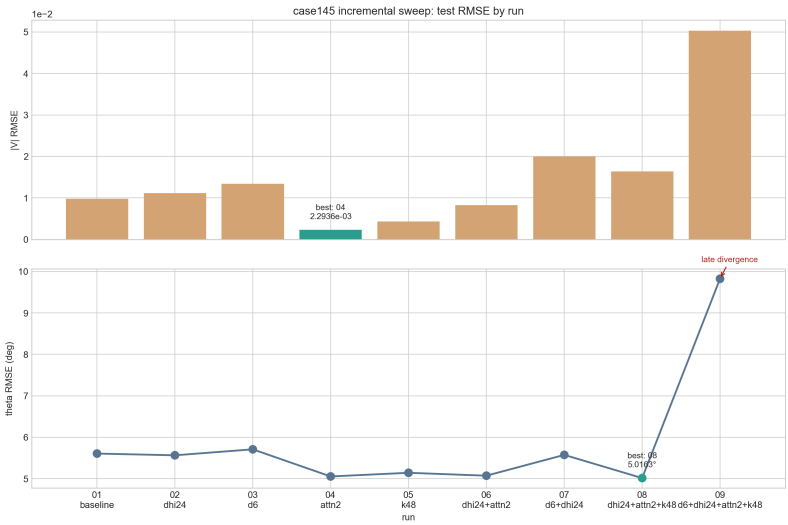


Figure 6: case145 incremental run summary (earlier small-config sweep). All configs plateau at $\theta \approx 5$–6°, well above the best from the full capacity sweep (1.77°).